

VALID: A 12-Item Validation Checklist for Financial Machine Learning

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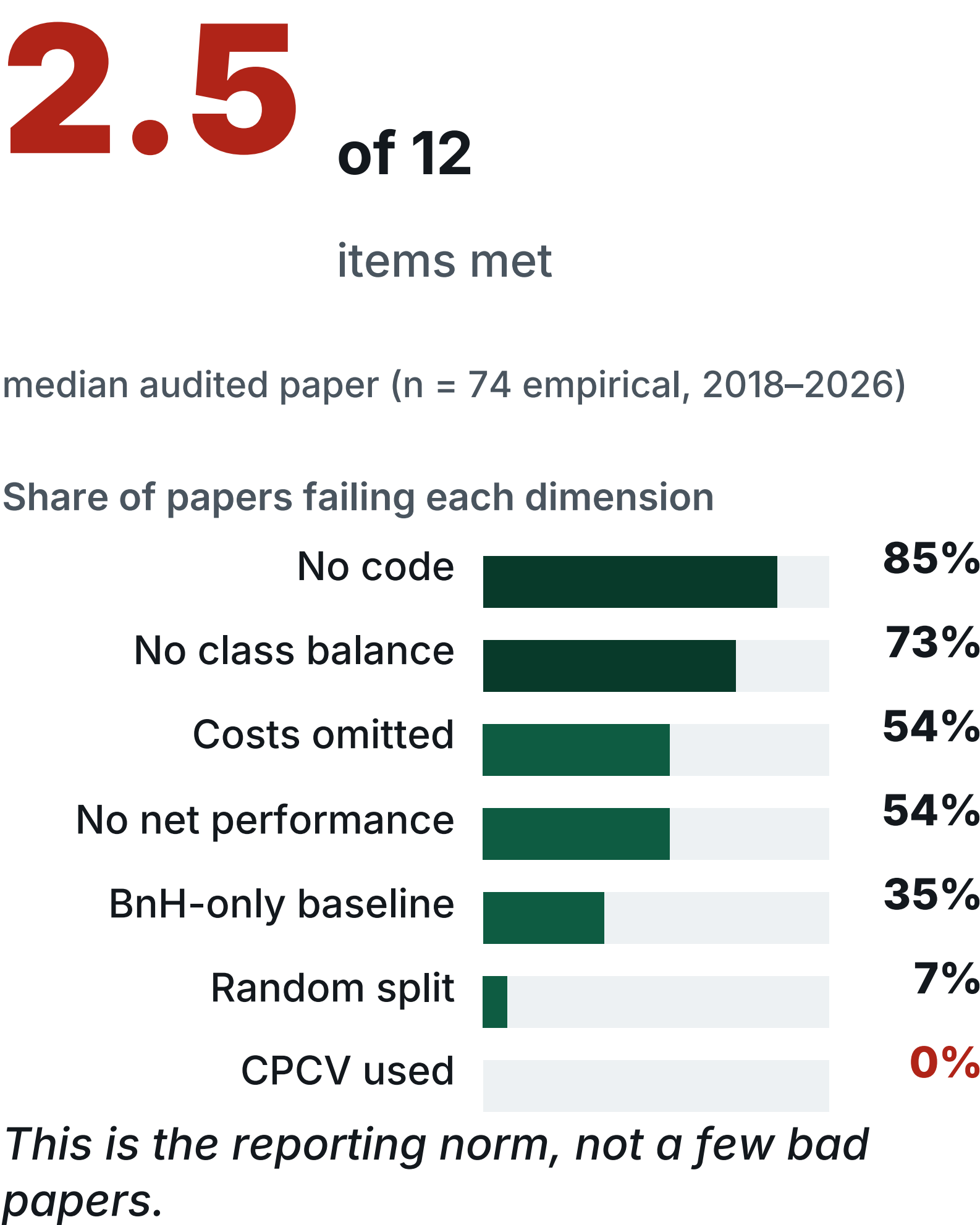


GitHub repo

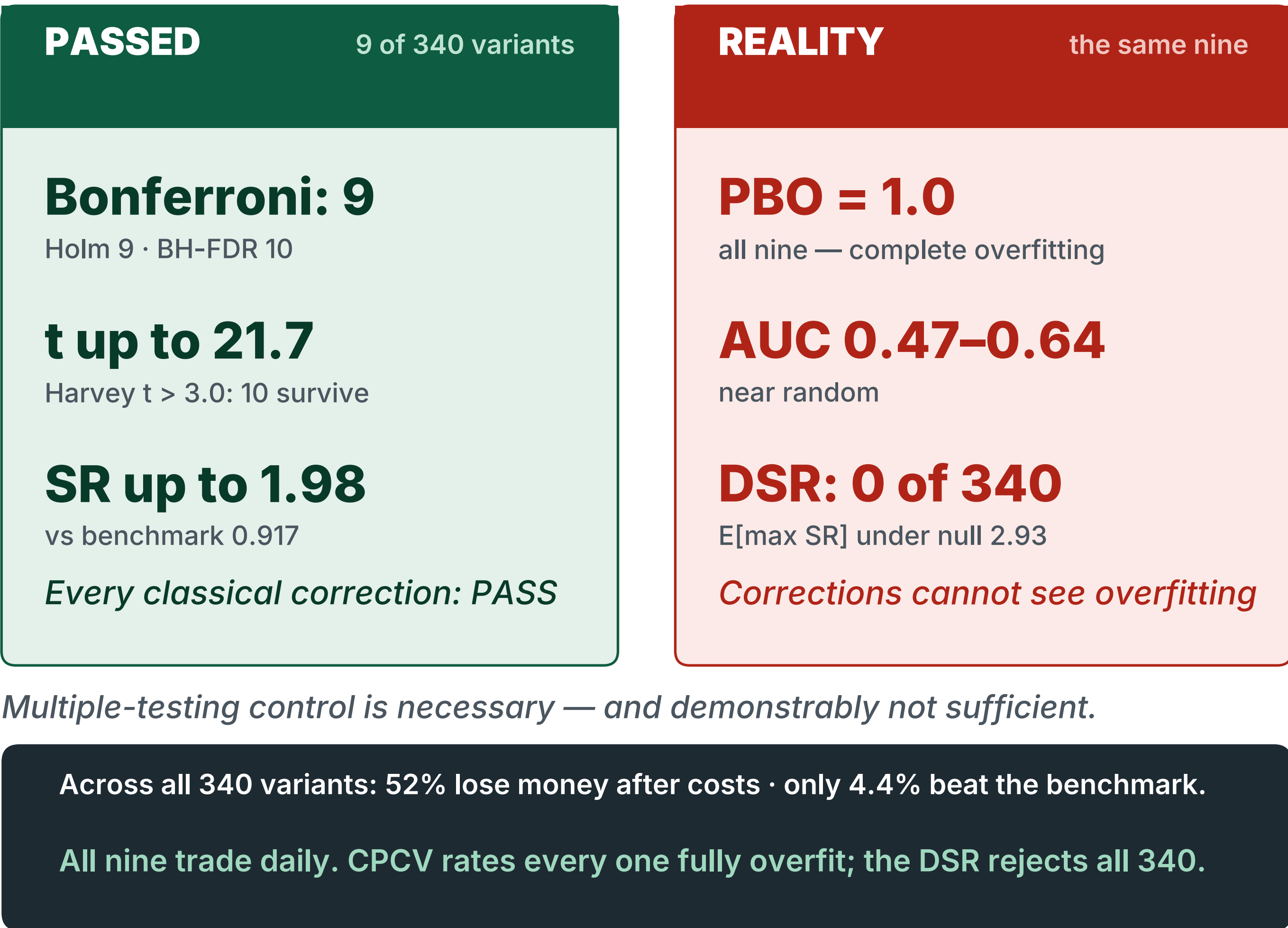
Statistically perfect.
Economically worthless.

340 strategy variants · 80 papers audited · one 12-item gate that tells them apart in ~5 minutes

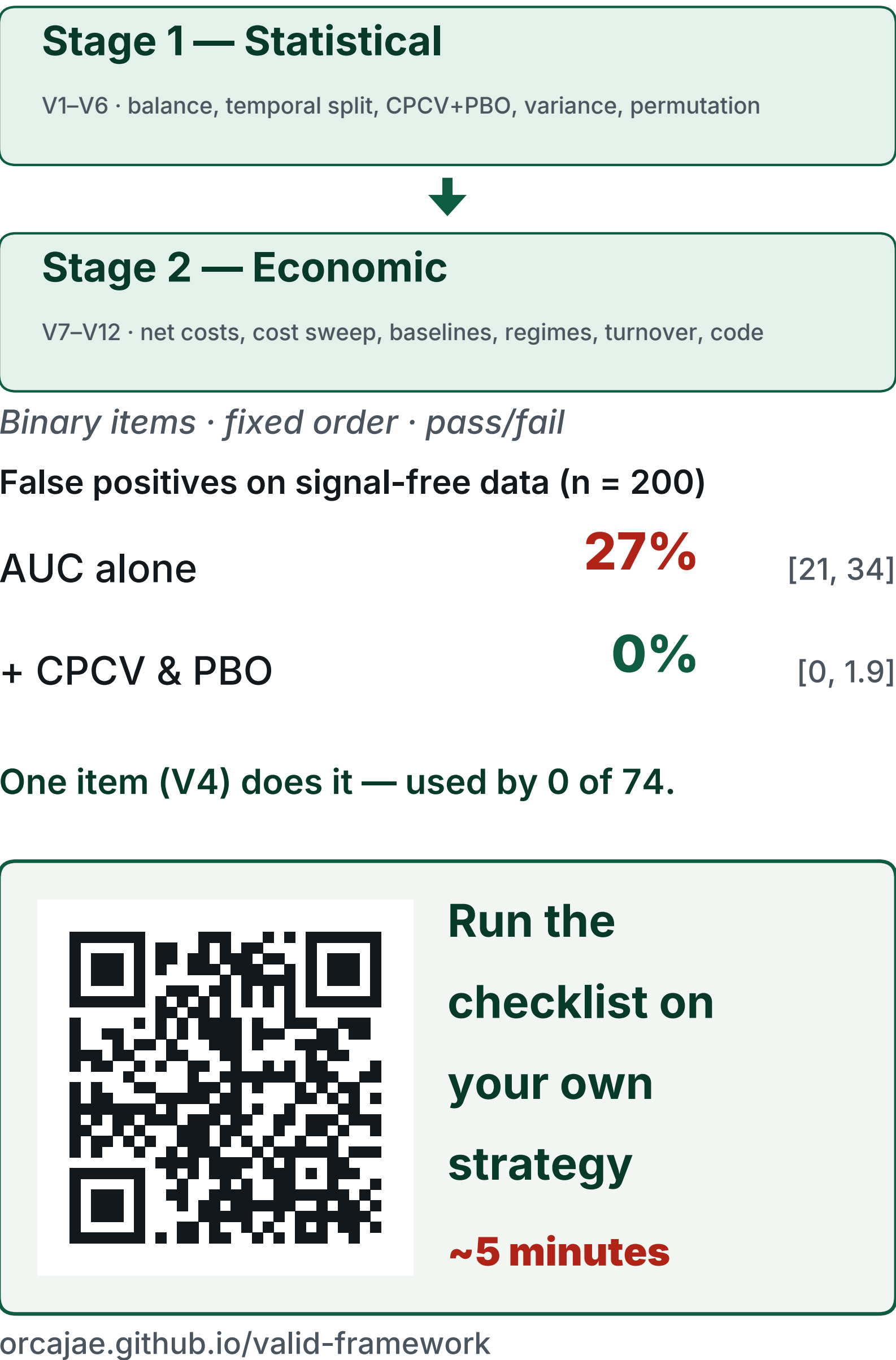
The Field Today



The Strategies That Passed Everything

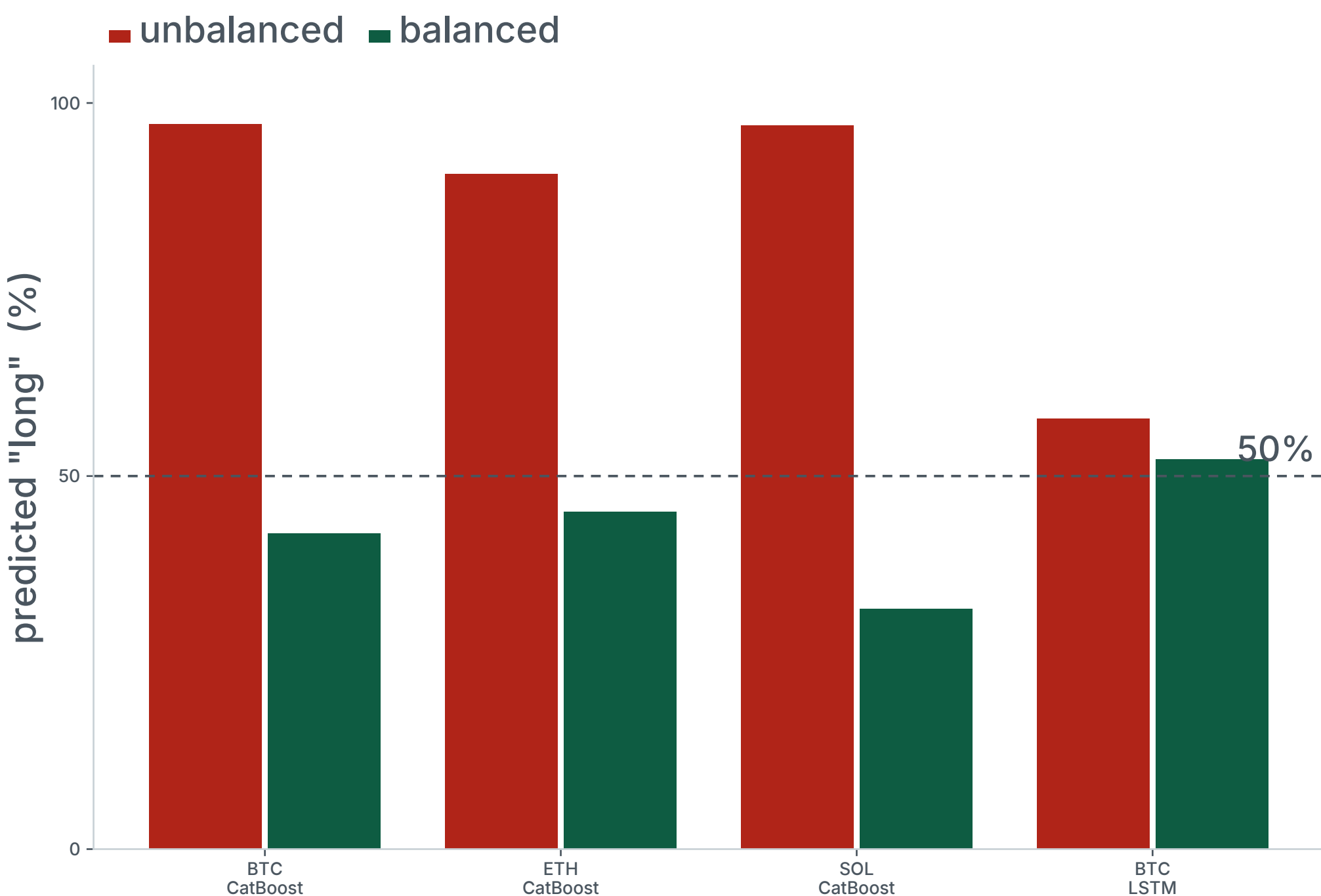


The Gate



Bull Bias

structural class imbalance (V1–V2)



Tree models predict "long" 90–97% of the time. After balancing, AUC converges to ≈0.50 — a coin flip.

Balanced AUC (BTC / ETH / SOL)	0.50 · 0.49 · 0.51
Net SR, 1h balanced	−0.88 · −1.28 · −0.21
18 combinations, BTC mean long	85%
BTC mean balanced AUC / net SR	0.484 / −0.174

All 1h models: PBO = 1.000.

Cost Illusion

net-of-cost performance by frequency (V7–V8, V11)



Net Sharpe falls monotonically with trading frequency. At 15 minutes every single variant loses money after 18 bp costs.

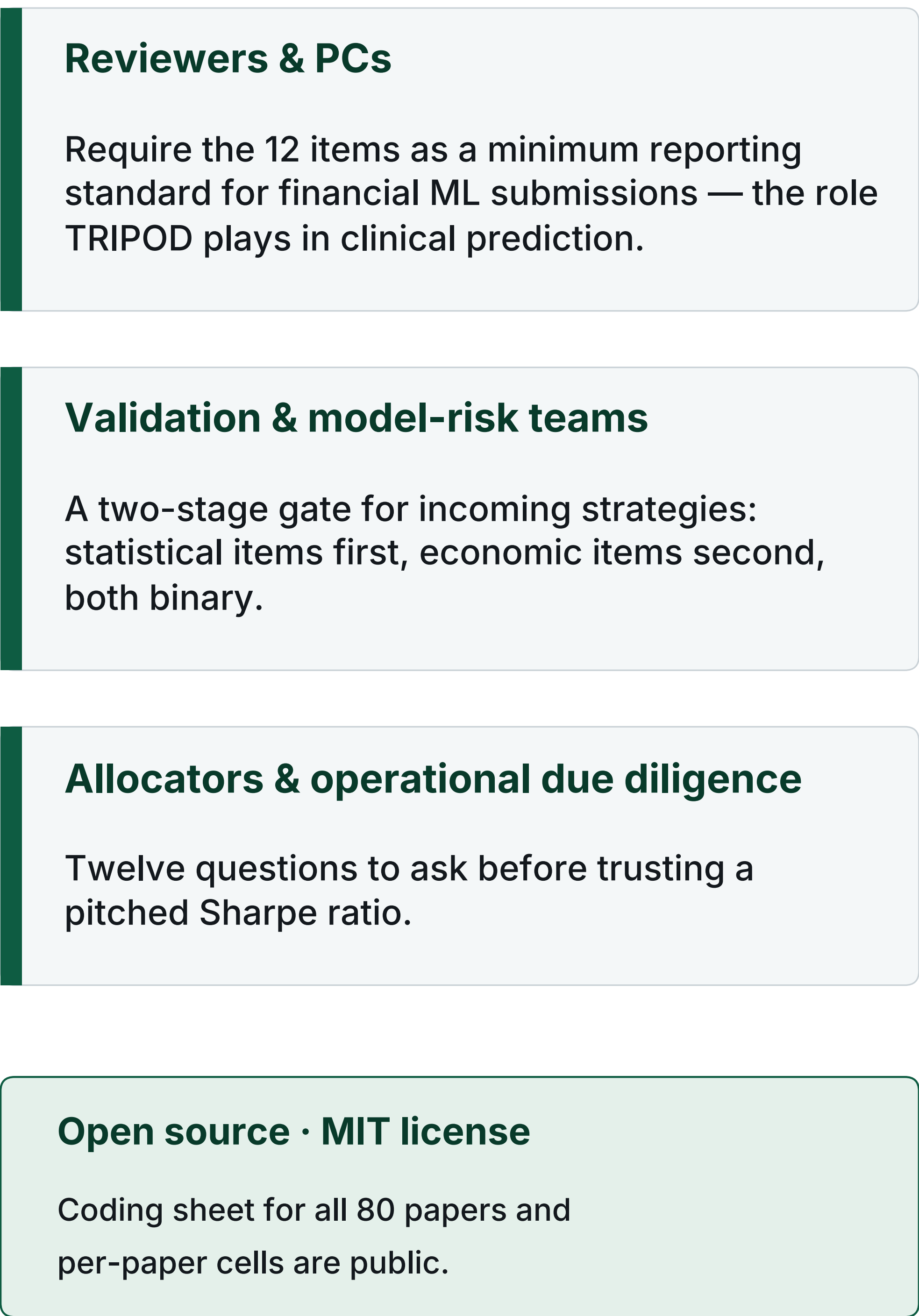
	median	best	< 0
15m (n = 9)	−1.80	−1.33	100%
1h (n = 133)	−1.32	+0.32	96%
4h (n = 9)	+0.28	+0.57	44%
1d (n = 77)	+0.54	+1.98	22%

One 1h CatBoost run: gross SR 0.750 → net 0.135 at 18 bp — 82% of gross Sharpe consumed at 101 trades / yr.

340-variant corpus, 18 bp round-trip, model variants only.

Who Is This For?

adoption paths



Score the last paper you reviewed

median in our audit: 2.5 / 12

- ☐ **V1** Report prediction class distribution
- ☐ **V2** Test with / without class balancing
- ☐ **V3** Use temporal splitting only
- ☐ **V4** Apply CPCV with PBO
- ☐ **V5** Report Var(SR_IS)
- ☐ **V6** Include permutation tests (≥100)
- ☐ **V7** Report net performance with costs
- ☐ **V8** Cost sensitivity analysis
- ☐ **V9** Compare against simple baselines
- ☐ **V10** Evaluate in bear markets and regime transitions
- ☐ **V11** Report trade frequency
- ☐ **V12** Provide code for reproducibility

V1–V6 Stage 1 (statistical) V7–V12 Stage 2 (economic)

v1.0 — not a certification. The checklist has had no formal Delphi review; community revision is invited.

Selected references

- Harvey, Liu & Zhu (2016). ...and the Cross-Section of Expected Returns. RFS 29(1).
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- Witzany (2021). A Bayesian Approach to Measurement of Backtest Overfitting. Risks 9(1).
- Kapoor & Narayanan (2023). Leakage and the Reproducibility Crisis in ML-based Science. Patterns 4(9).